

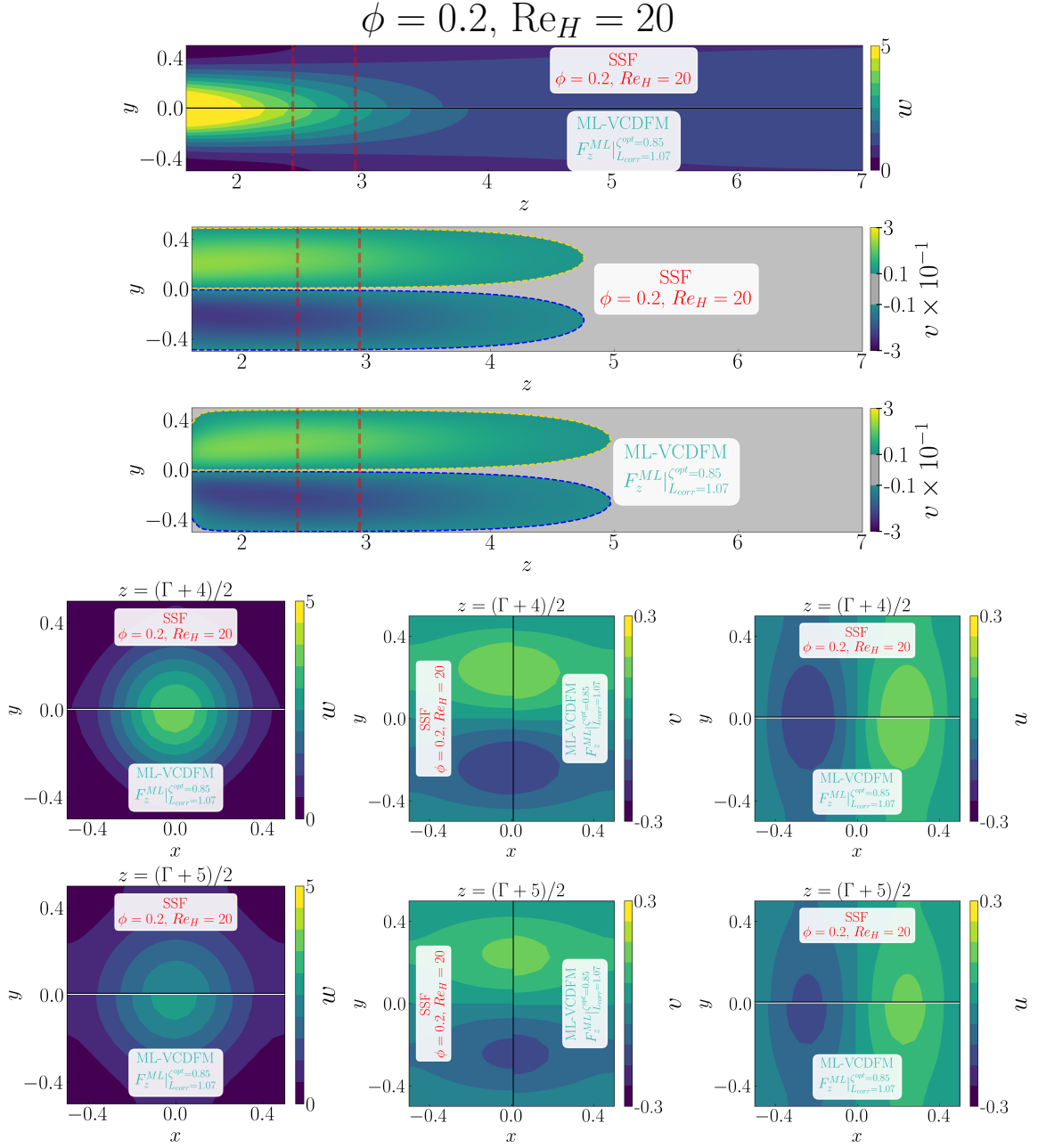


Figure D.146: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 20$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 4)/2$ in the first row and $z = (\Gamma + 5)/2$ in the second row.

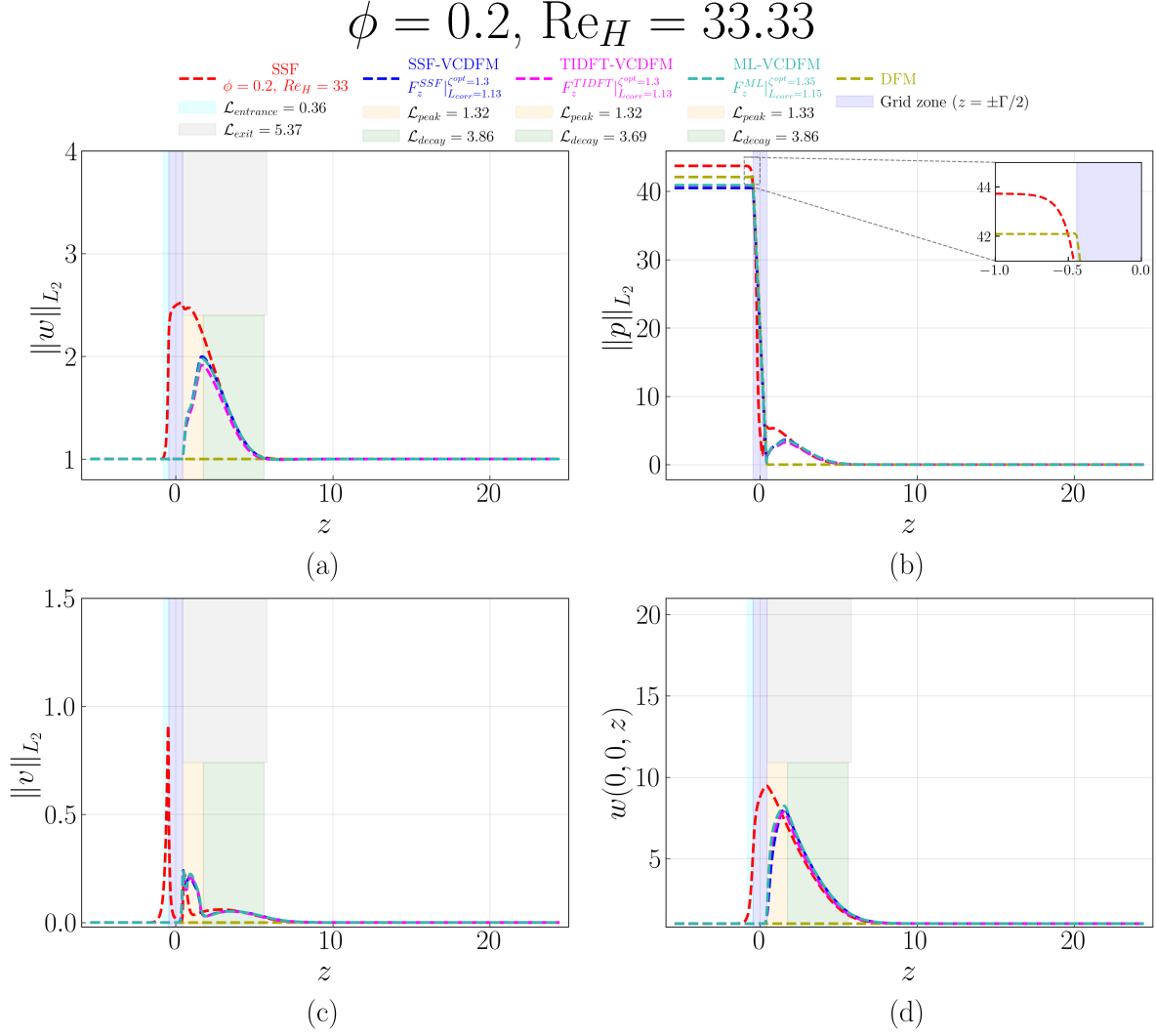


Figure D.147: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 33.33$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

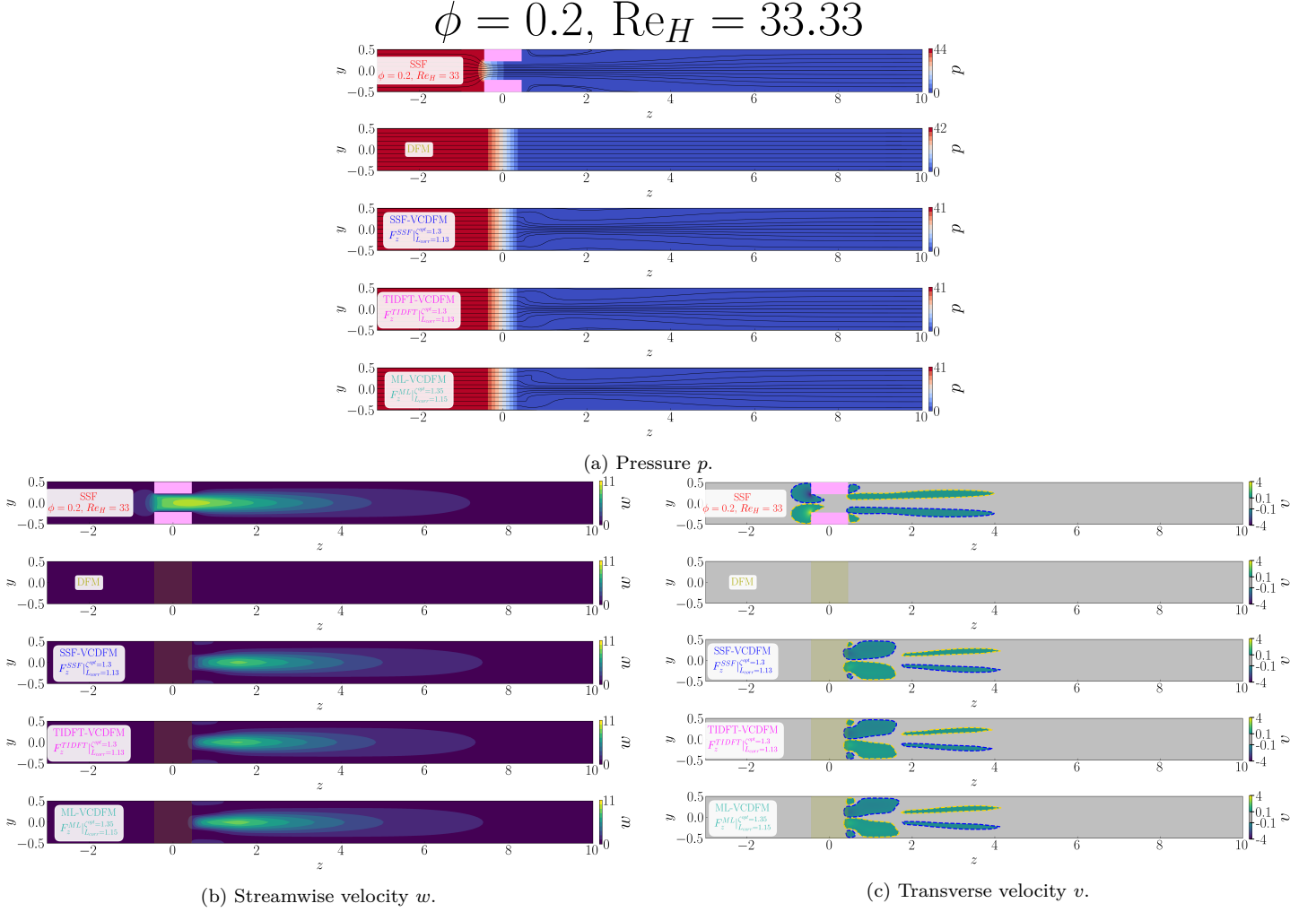


Figure D.148: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.2$, $\text{Re}_H = 33.33$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

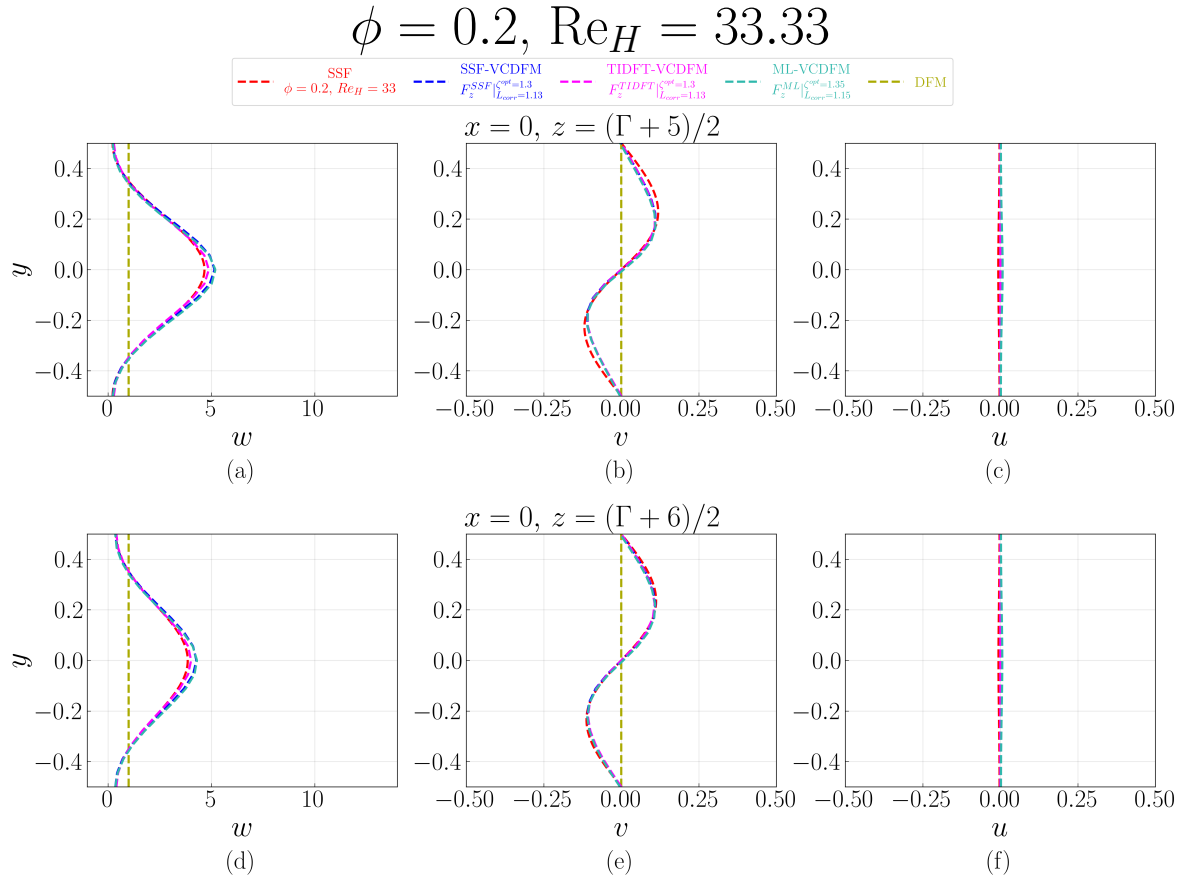


Figure D.149: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 33.33$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 5)/2$ in (a–c) and $z = (\Gamma + 6)/2$ in (d–f).

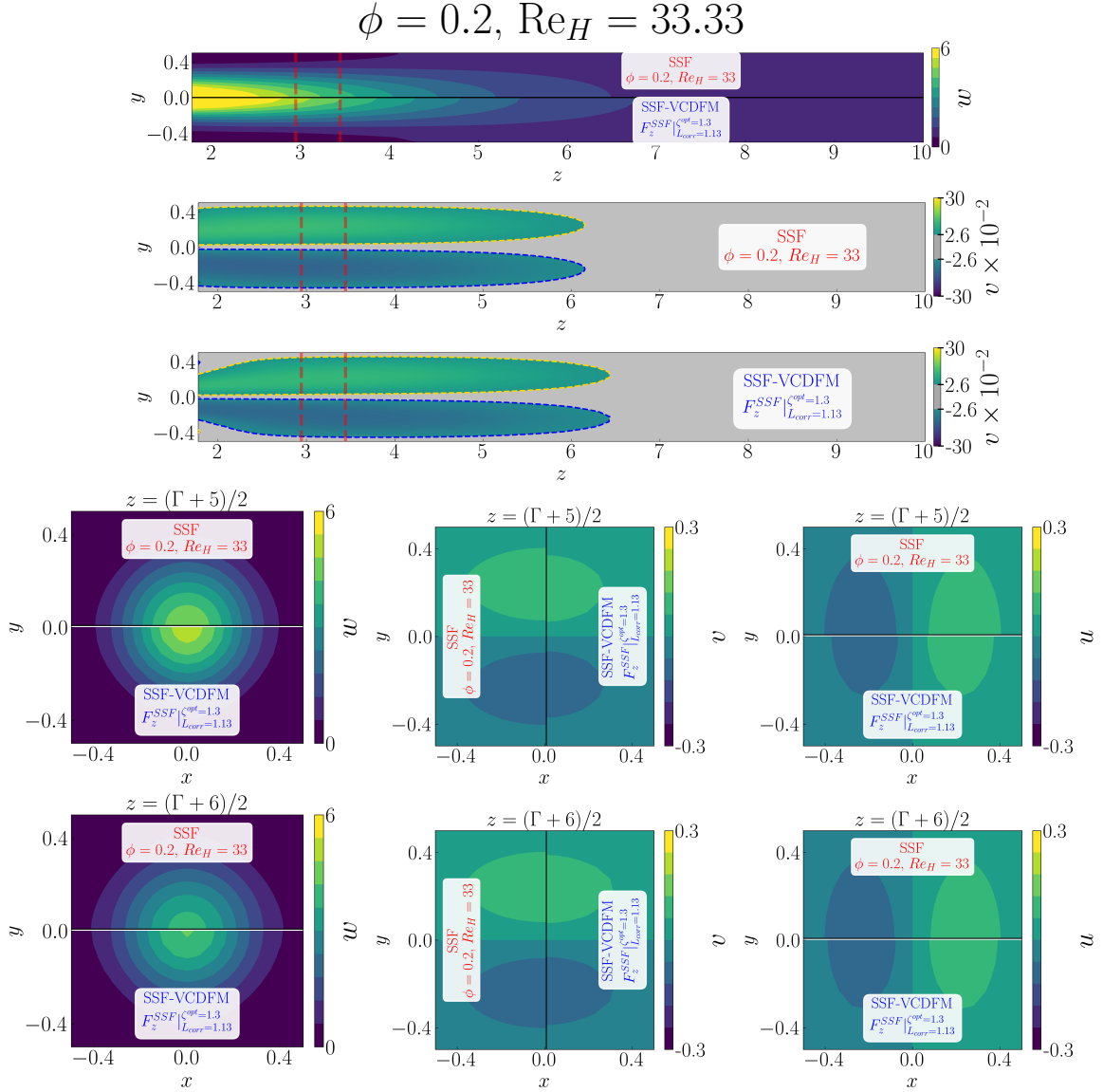


Figure D.150: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.2$ and $Re_H = 33.33$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 5)/2$ in the first row and $z = (\Gamma + 6)/2$ in the second row.

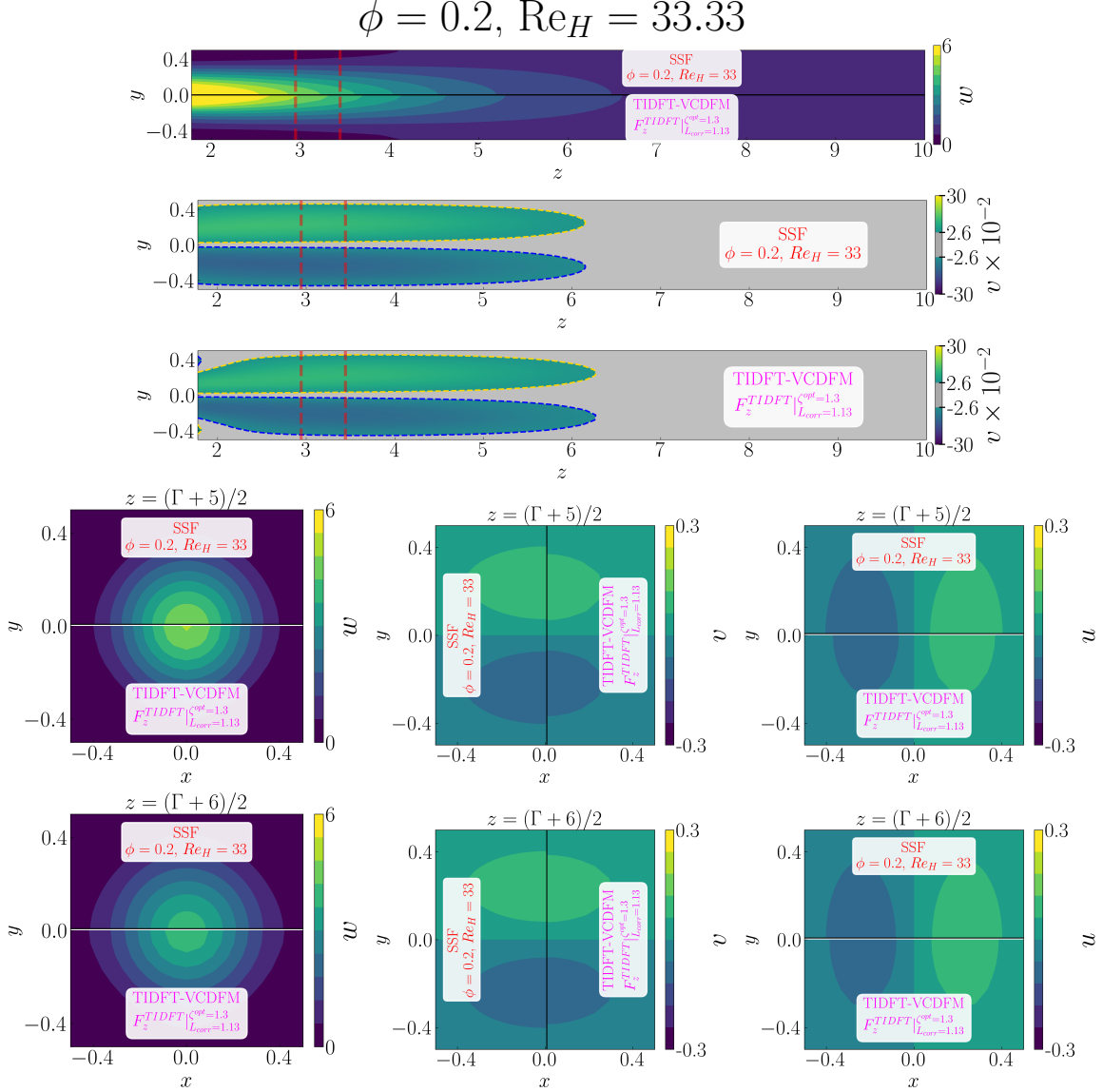


Figure D.151: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.2$ and $Re_H = 33.33$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 5)/2$ in the first row and $z = (\Gamma + 6)/2$ in the second row.

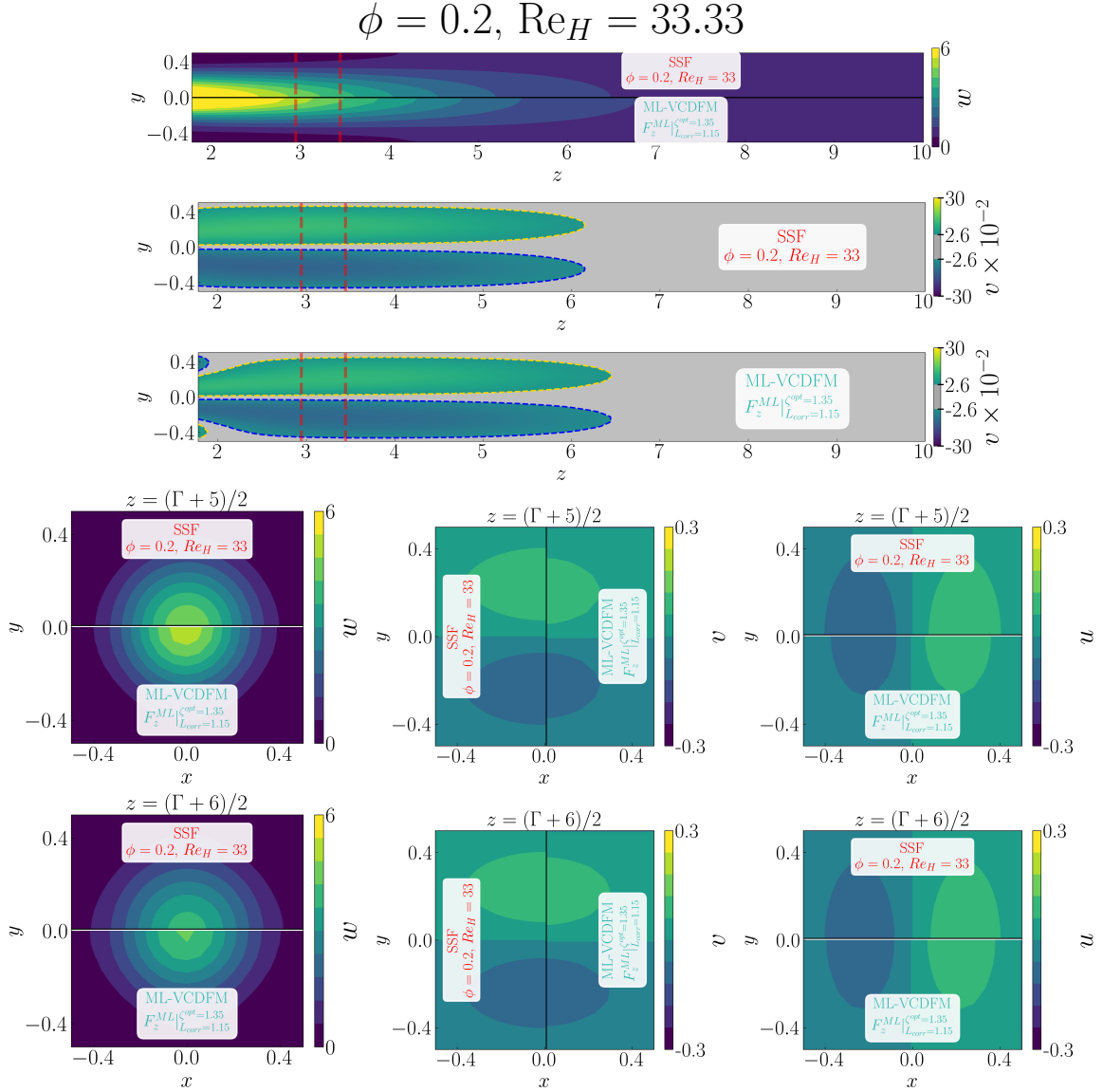


Figure D.152: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 33.33$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 5)/2$ in the first row and $z = (\Gamma + 6)/2$ in the second row.

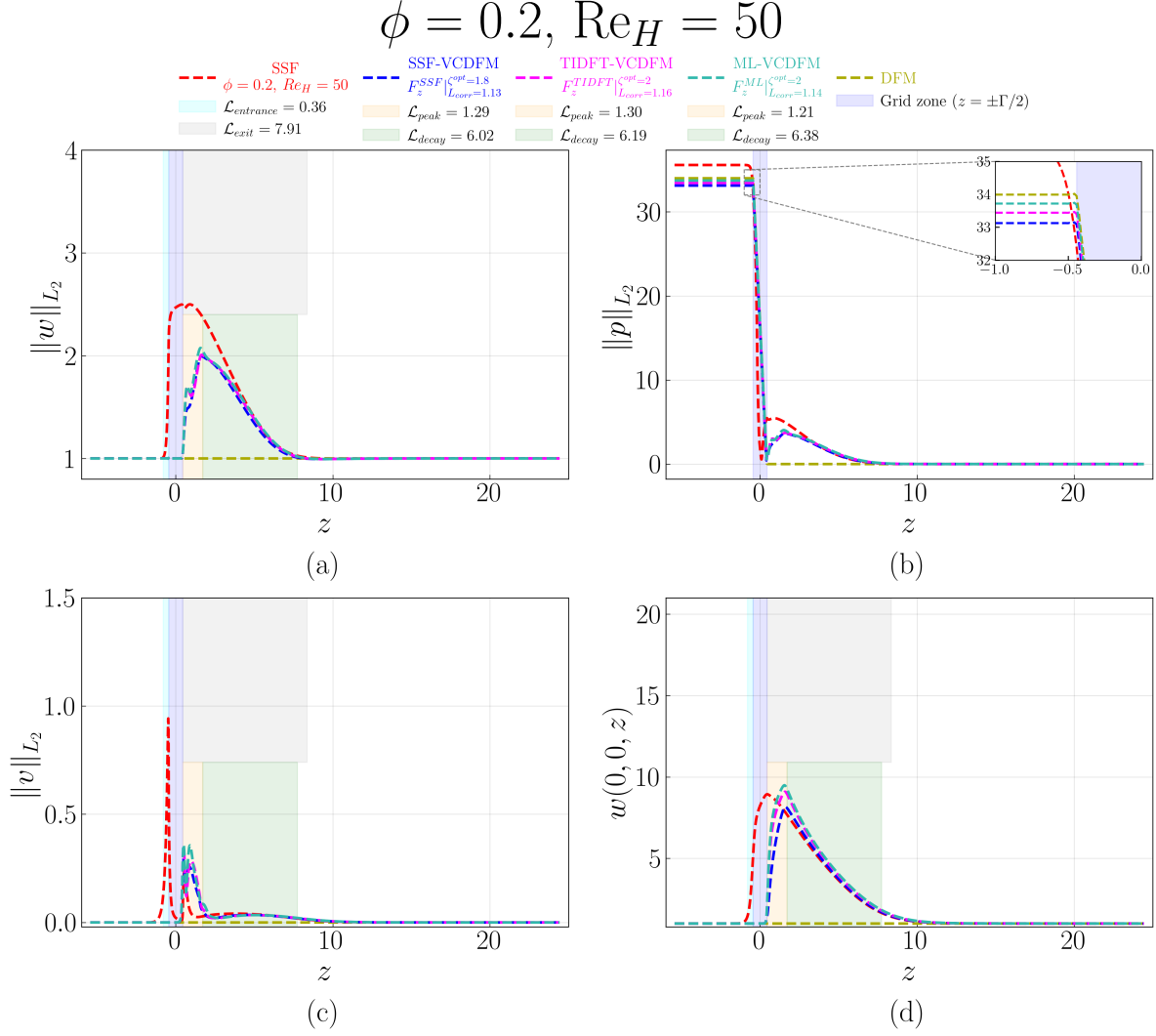


Figure D.153: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 50$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based L_{peak} and L_{decay} are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

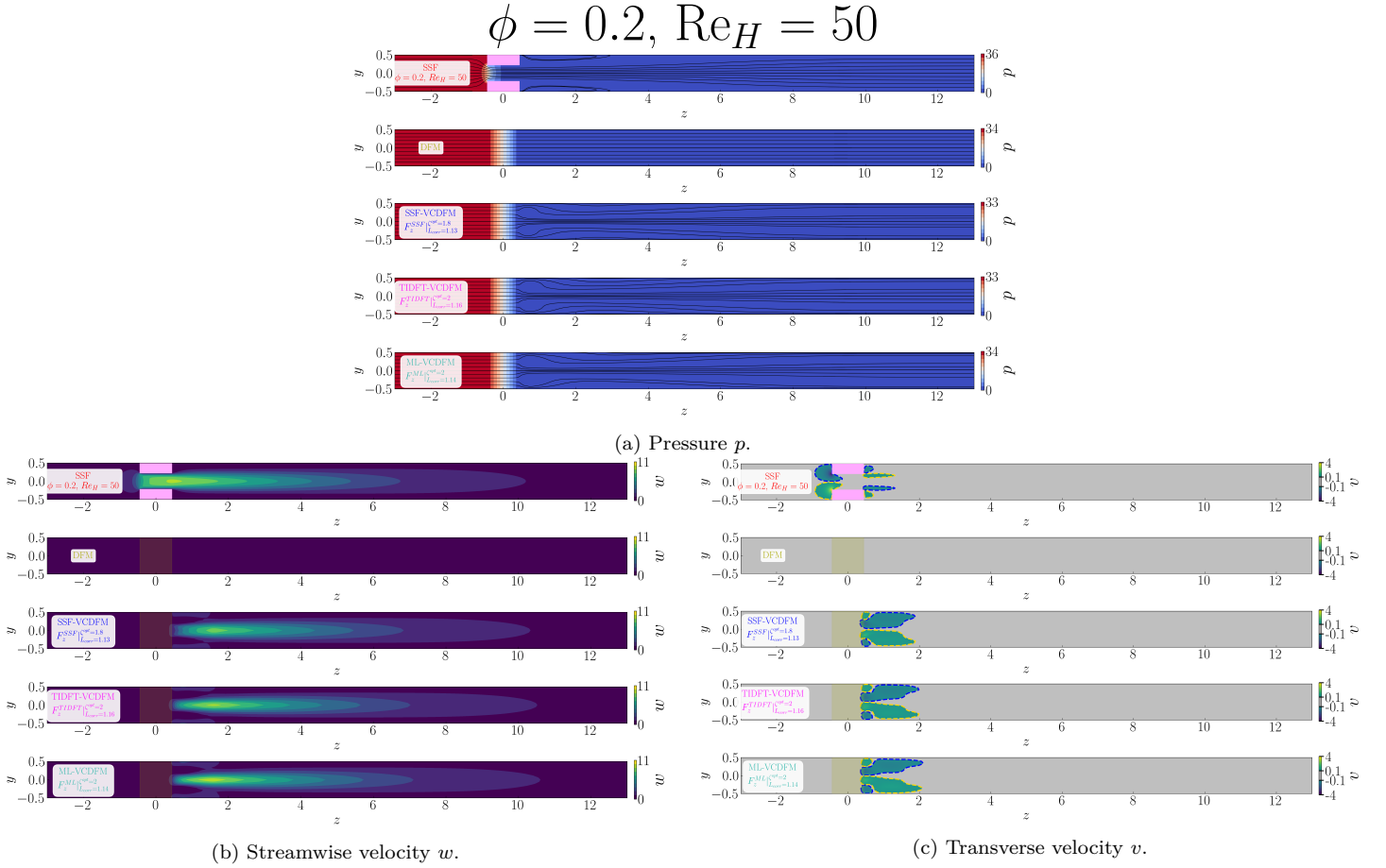


Figure D.154: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.2$, $\text{Re}_H = 50$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

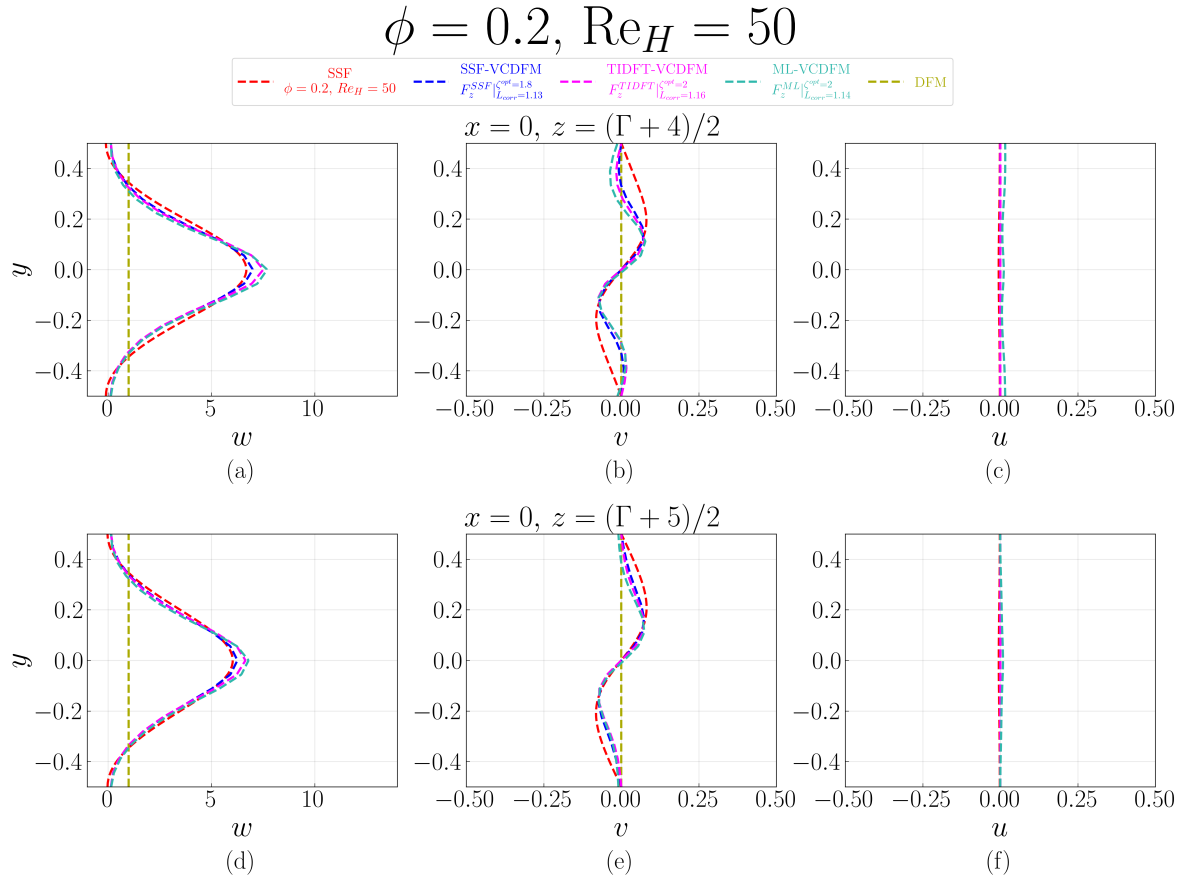


Figure D.155: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 50$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (2\Gamma + 9)/4$ in (a–c) and $z = (2\Gamma + 11)/4$ in (d–f).

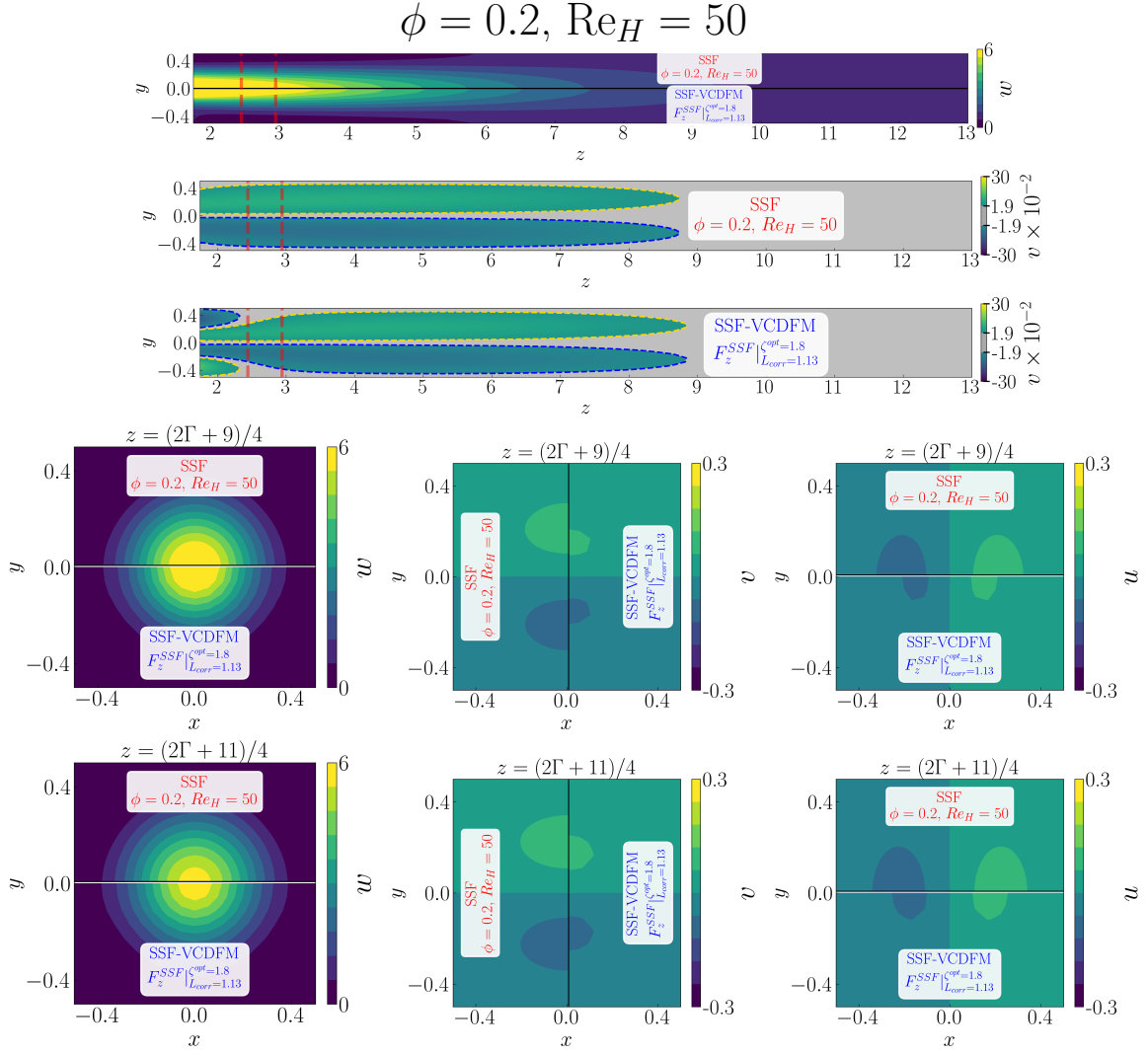


Figure D.156: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 50$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (2\Gamma + 9)/4$ in the first row and $z = (2\Gamma + 11)/4$ in the second row.

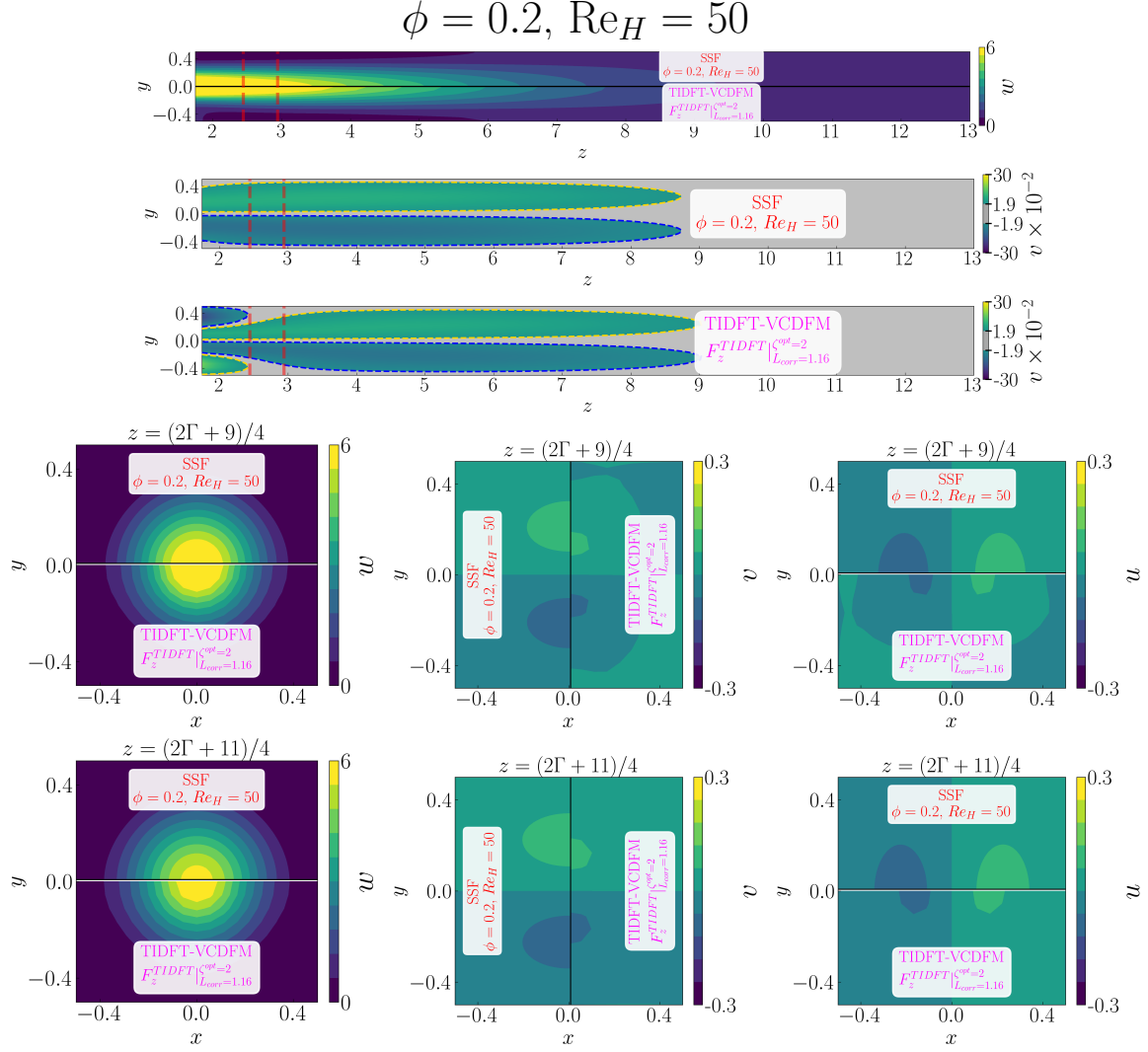


Figure D.157: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.2$ and $Re_H = 50$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (2\Gamma + 9)/4$ in the first row and $z = (2\Gamma + 11)/4$ in the second row.

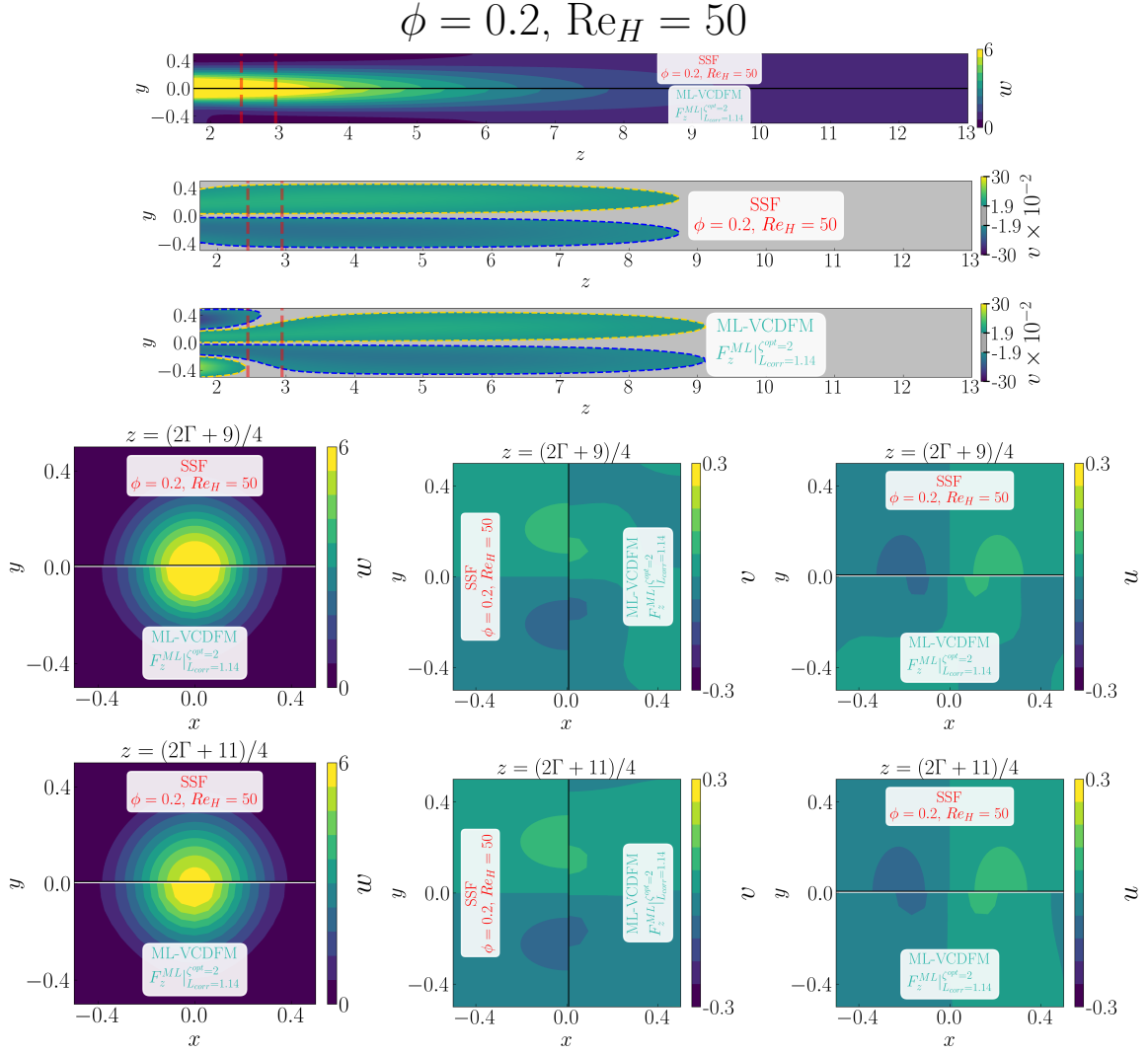


Figure D.158: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.2$ and $\text{Re}_H = 50$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (2\Gamma + 9)/4$ in the first row and $z = (2\Gamma + 11)/4$ in the second row.

$$\phi = 0.25, \text{Re}_H = 0.05$$

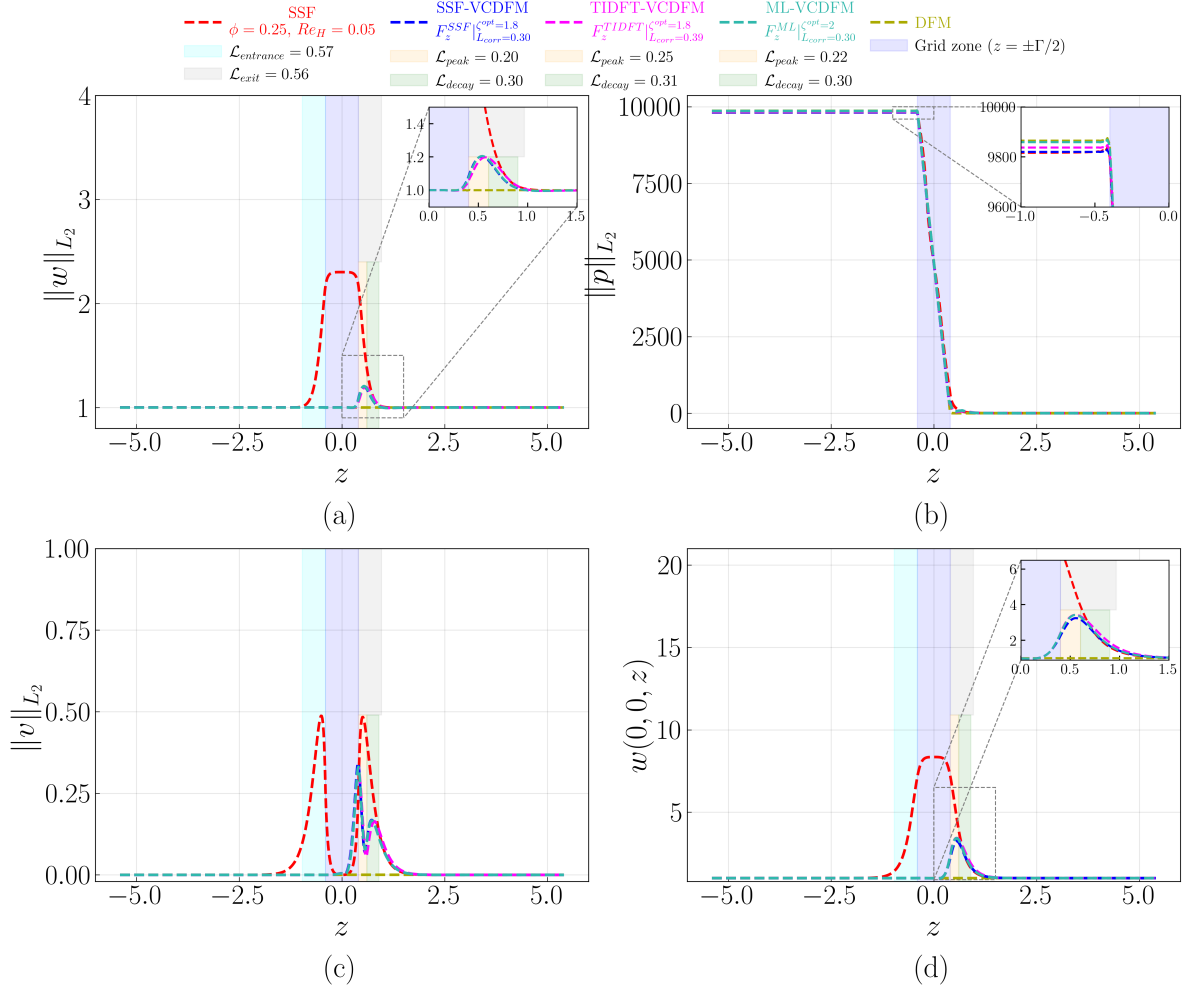


Figure D.159: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 0.05$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

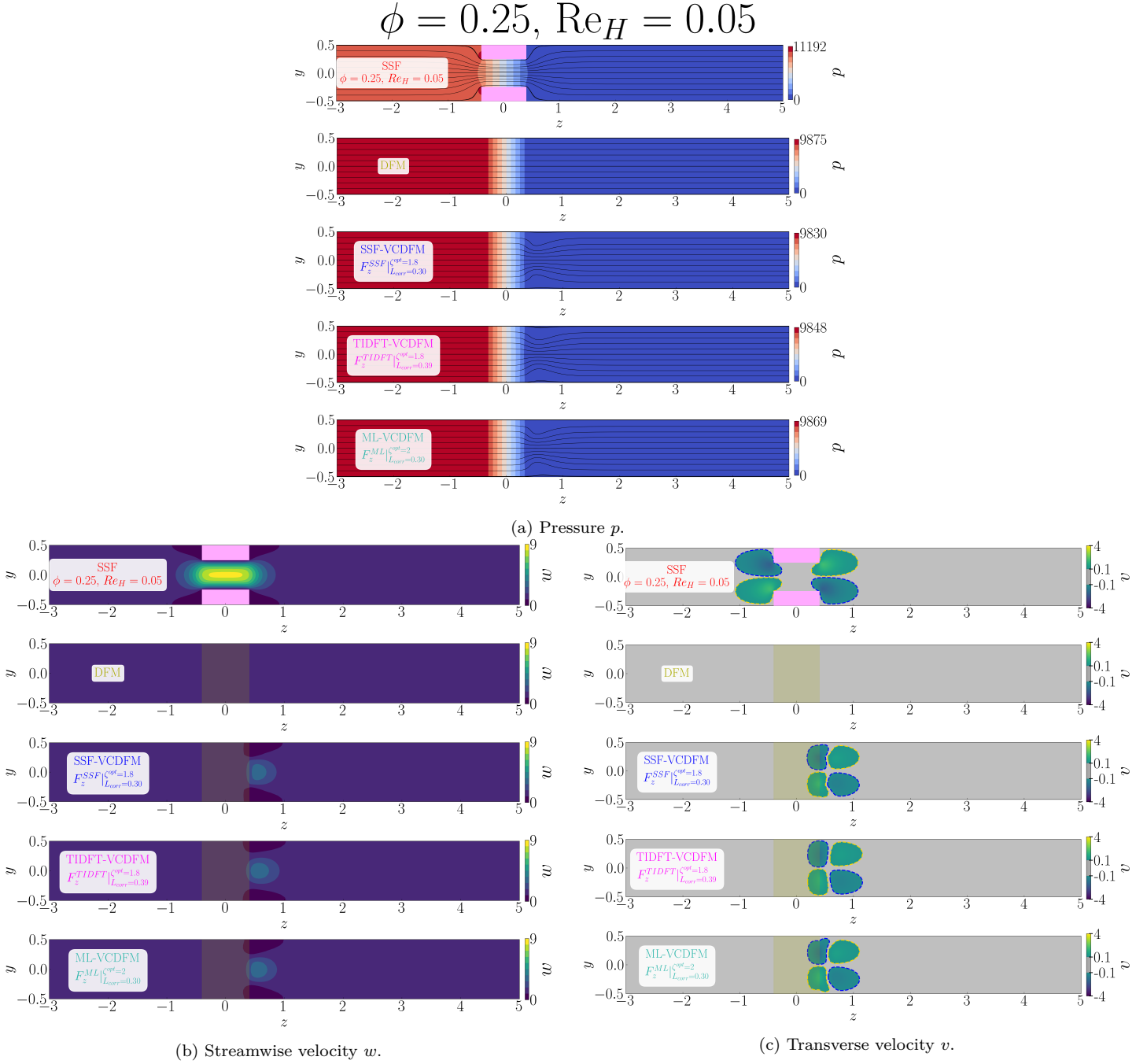


Figure D.160: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.25$, $Re_H = 0.05$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

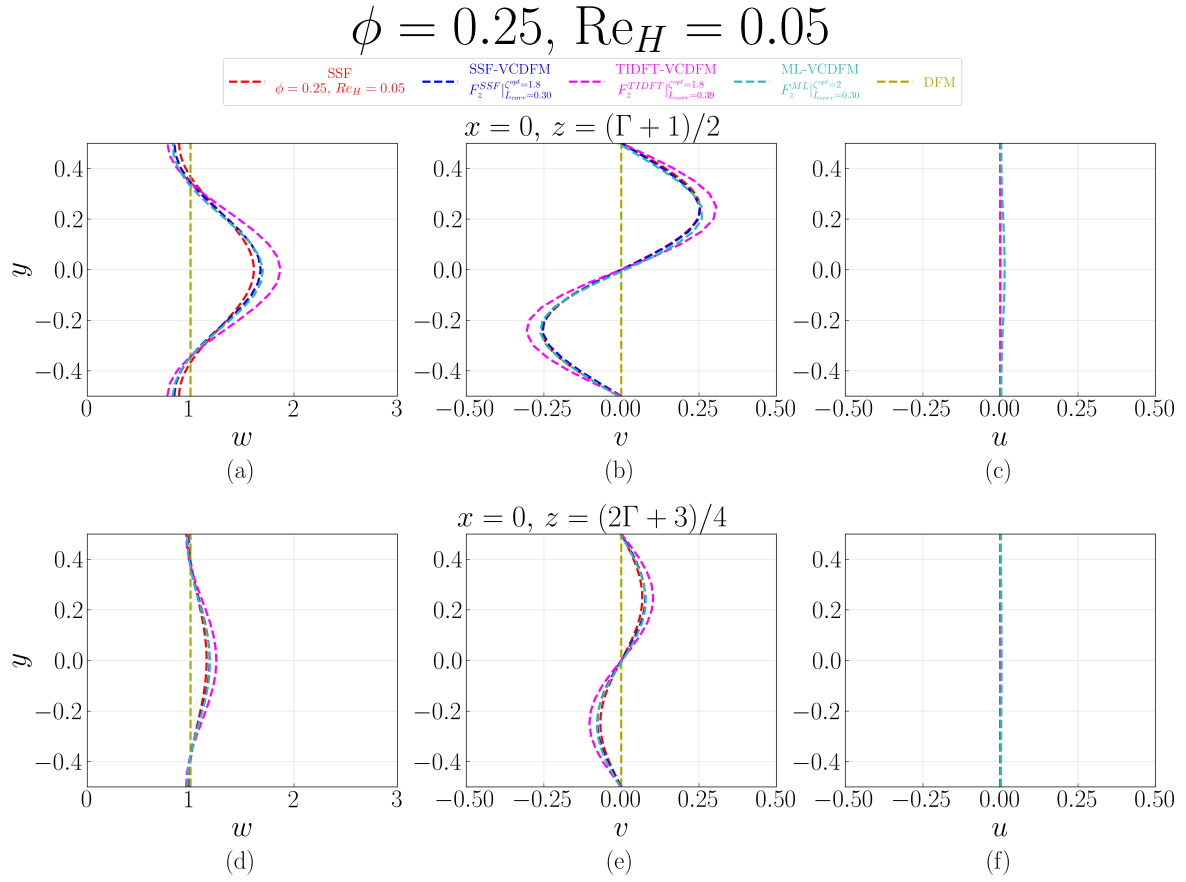


Figure D.161: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 0.05$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 1)/2$ in (a–c) and $z = (2\Gamma + 3)/4$ in (d–f).

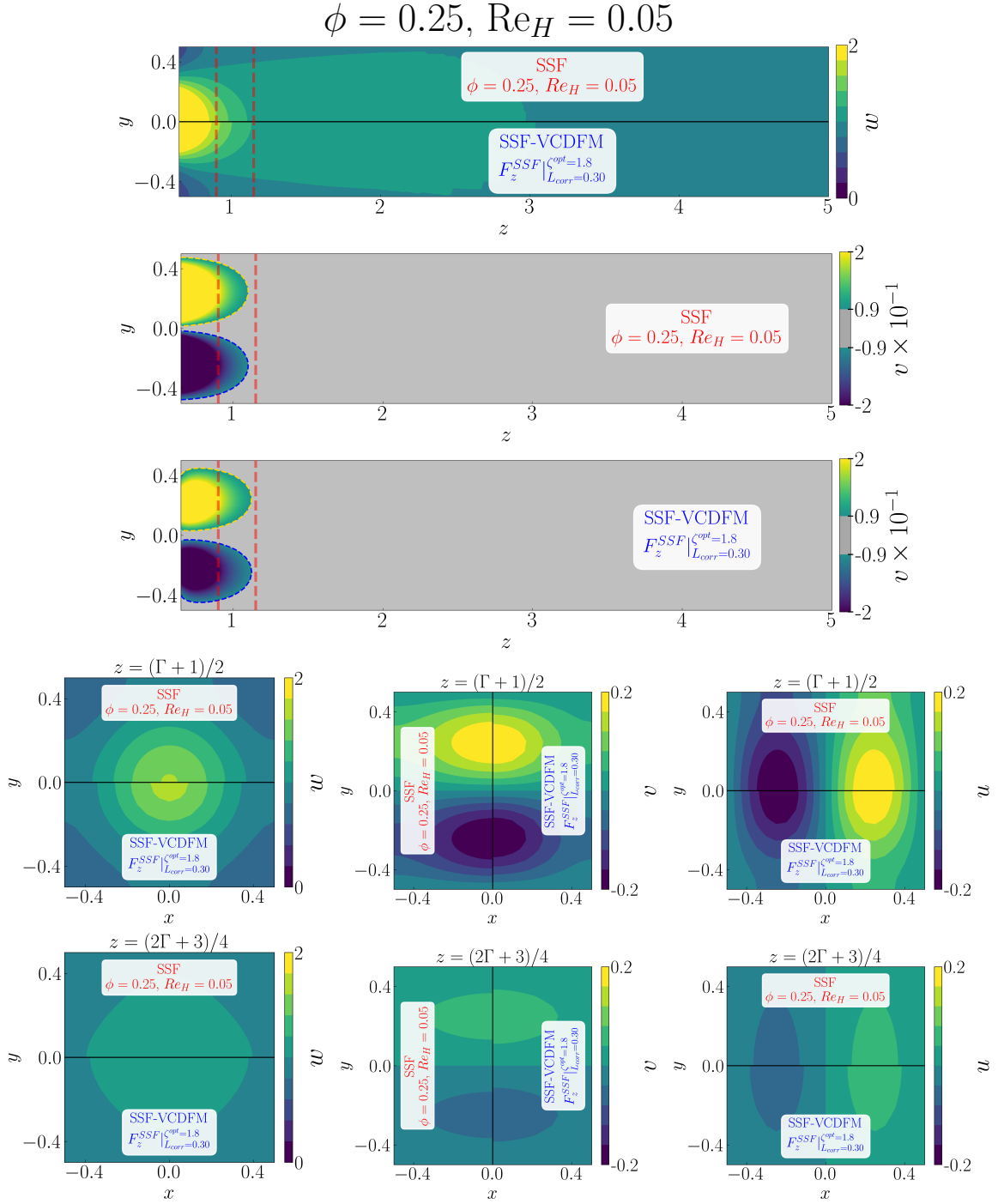


Figure D.162: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.25$ and $Re_H = 0.05$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

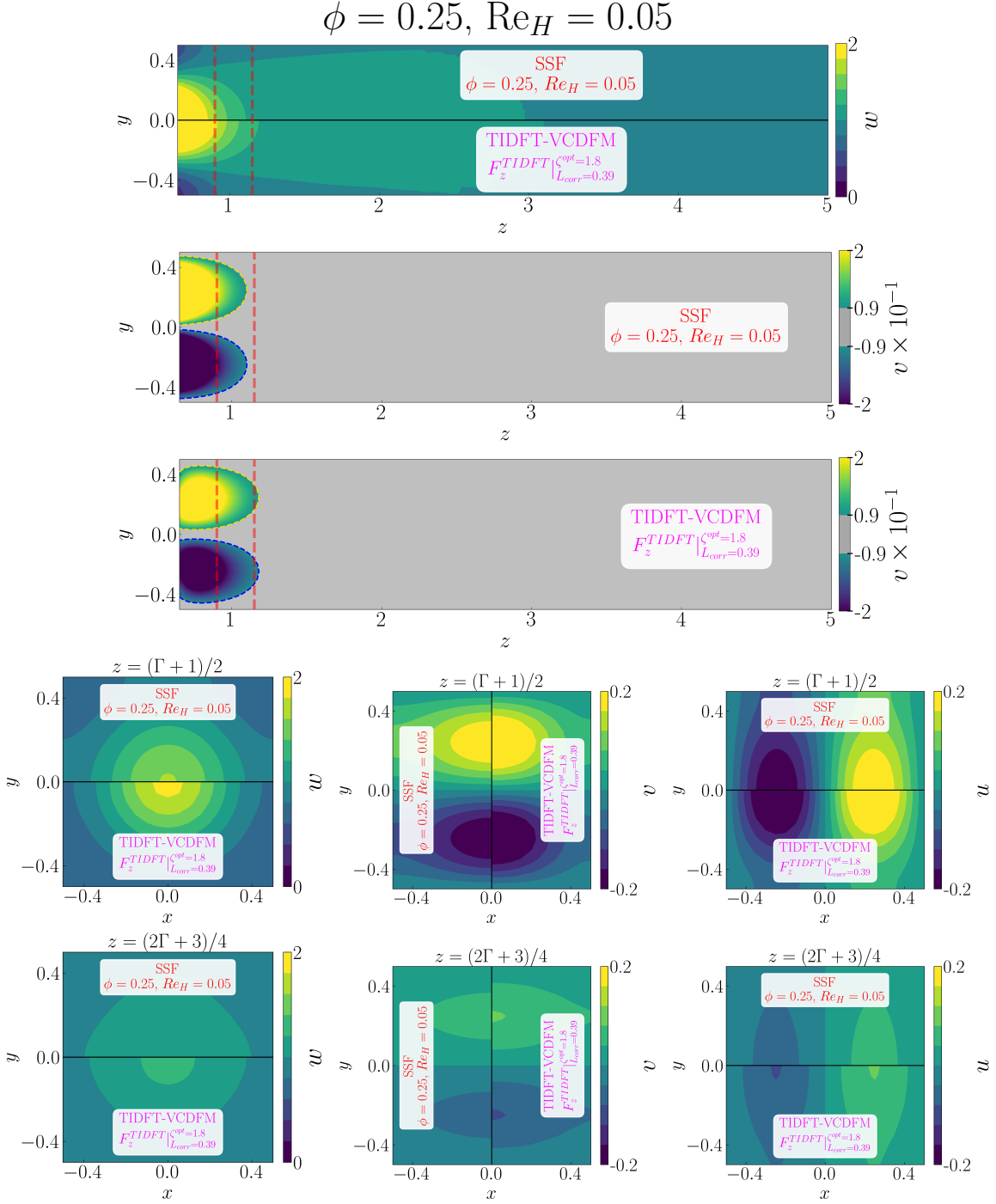


Figure D.163: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 0.05$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

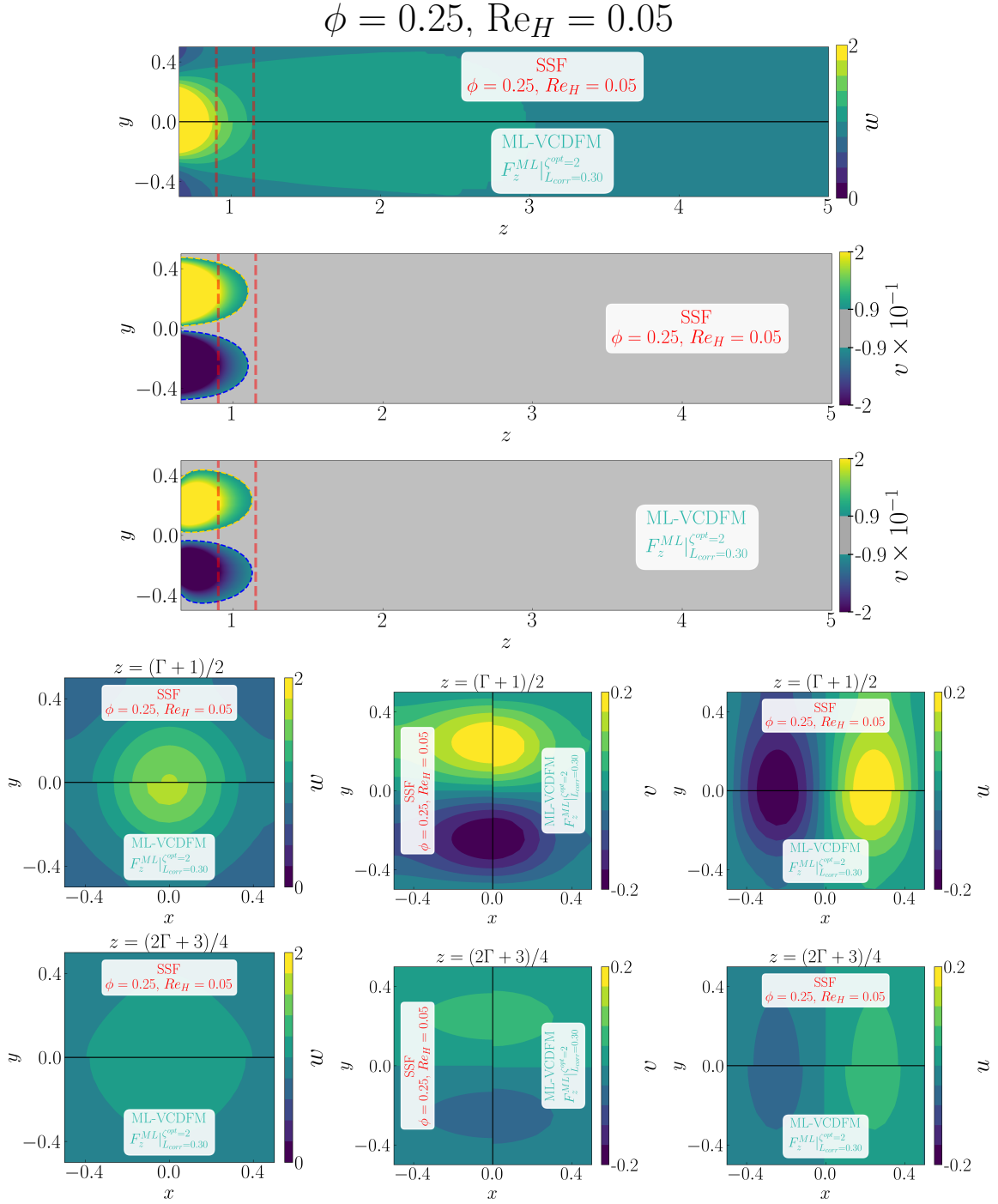


Figure D.164: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.25$ and $Re_H = 0.05$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

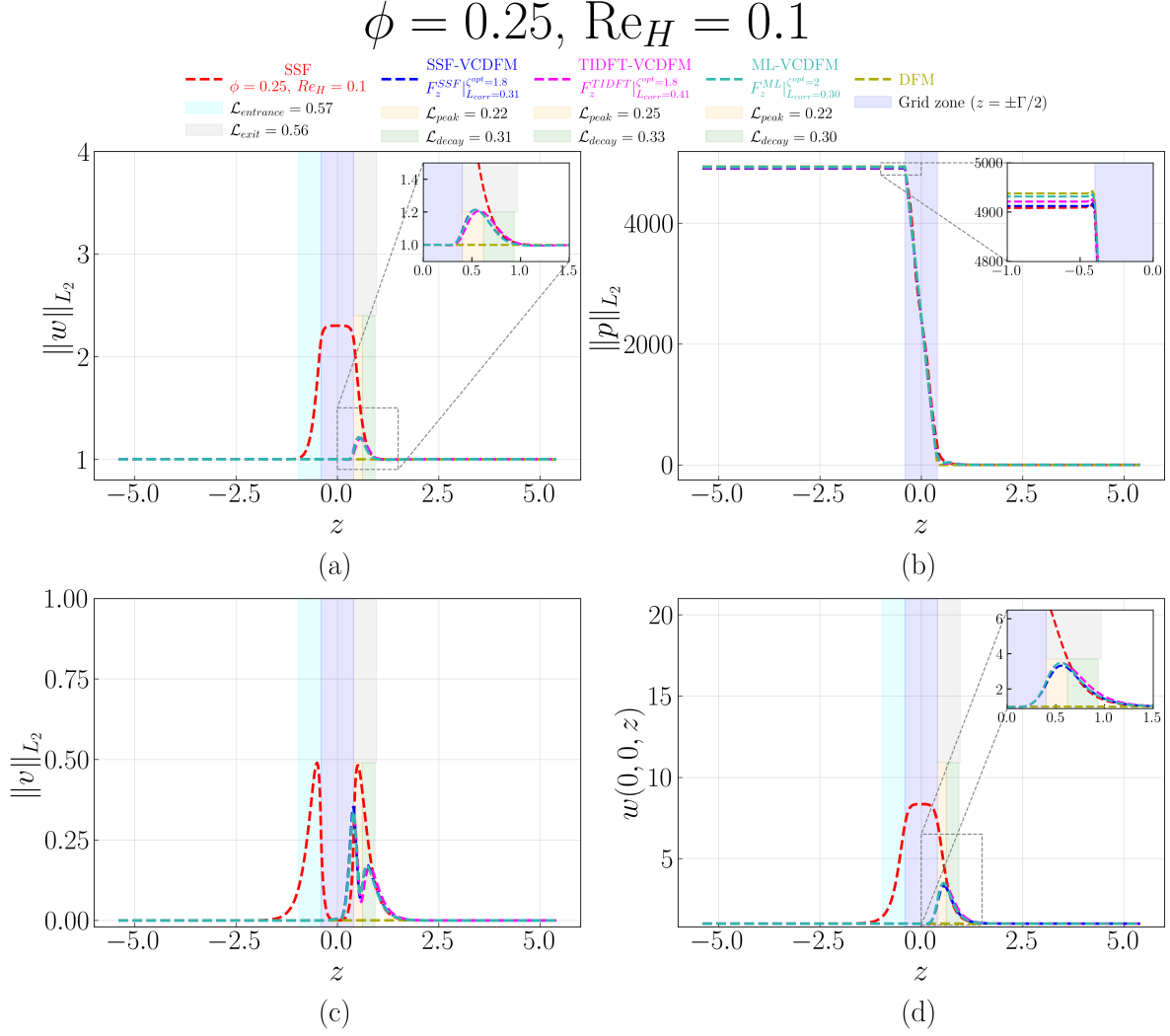


Figure D.165: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.25$ and $\text{Re}_H = 0.1$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.